

Rural Employment Analytics Dashboard Using Power BI: A Data-Driven Approach to MGNREGA Performance Monitoring

Dhruv Kumar

School of Computer Science and Engineering
Lovely Professional University, Punjab, India

Registration No.: 12409374 | Roll No.: 57

Abstract

The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) generates vast volumes of administrative data across millions of rural households in India, encompassing employment generation, expenditure patterns, and work progress metrics. Despite the availability of this data, the absence of an integrated analytical framework limits the ability of policymakers and administrators to derive timely and actionable insights. This paper presents the design, development, and evaluation of a comprehensive multi-page interactive analytics dashboard using Microsoft Power BI for monitoring and evaluating MGNREGA performance at national, state, and district levels. The system incorporates data preprocessing through Power Query, DAX-based calculated measures, and multiple visualization types including KPI cards, choropleth maps, stacked bar charts, treemaps, and area charts. Six analytical objectives are addressed: work progress monitoring, state-wise distribution analysis, expenditure breakdown, employment and labour participation analysis, district-level performance evaluation, and an executive KPI summary. The dashboard transforms raw MGNREGA CSV data into structured visual insights highlighting regional disparities, expenditure allocation patterns, employment utilization gaps, and work completion trends. Results demonstrate the efficacy of business intelligence tools in public governance monitoring and support evidence-based decision-making for rural development programs.

Keywords—MGNREGA; Power BI; Business Intelligence; Data Analytics; Rural Employment; Dashboard Design; Data Visualization; Public Governance

I. INTRODUCTION

The rural economy of India is substantially dependent on government-supported livelihood programs, among which the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) stands as one of the most significant social welfare initiatives. The scheme generates extensive administrative and operational data every day—spanning job card registrations, work demand applications, wage payments, and asset creation records across thousands of Gram Panchayats nationwide.

The importance of structured data management and analytics for MGNREGA has grown substantially, given the scale of implementation covering millions of households across multiple administrative levels including villages, blocks, districts, and states. Data analytics enables a shift from traditional paperwork-based monitoring systems to a more dynamic, transparent, and accountable governance model [1].

By analyzing historical and real-time data, administrators can identify patterns such as seasonal spikes in employment demand, delays in wage payments, and underutilization of funds in specific regions—enabling proactive decision-making and better resource allocation. Dashboards as graphical interfaces present key indicators in a single interactive view, offering dynamic filtering, near

real-time updates, and visual representations that facilitate comprehension of complex multi-dimensional data [2].

Microsoft Power BI, employed in this project, provides powerful data visualization capabilities, interactive features, and efficient handling of large datasets, making it well-suited for analyzing and presenting MGNREGA data in a structured manner. The primary purpose of this project is to design and develop a fully functional multi-page dashboard that transforms raw MGNREGA data into meaningful insights supporting better understanding, analysis, and decision-making.

II. BACKGROUND AND RELATED WORK

MGNREGA, enacted in 2005 under the Ministry of Rural Development, Government of India, guarantees 100 days of wage employment per financial year to rural households. The scheme is administered through a web-based MIS portal that aggregates data from Gram Panchayats up to the national level [3]. Despite rich data availability, meaningful multi-level analysis has remained challenging due to fragmented formats and the absence of integrated visualization frameworks.

Business intelligence tools such as Microsoft Power BI, Tableau, and QlikSense have been widely adopted for transforming raw organizational data into actionable insights [9][10]. In public governance and social welfare

domains, data-driven approaches have demonstrated measurable value in improving transparency, monitoring performance, and identifying systemic inefficiencies [5]. While predictive analytics integration with BI tools has been explored in related literature [12][13], dedicated multi-objective dashboard solutions for MGNREGA specifically remain limited, underscoring the contribution of this work.

III. PROBLEM STATEMENT

MGNREGA generates large volumes of data related to employment, expenditure, and workforce participation across multiple states and districts in India. However, much of this data is stored in raw tabular formats such as CSV and Excel files, making it difficult for policymakers and administrators to derive meaningful insights in a timely and efficient manner.

Despite the availability of rich data, the absence of a structured analytical and visualization system creates several critical challenges in monitoring and evaluating the performance of the scheme. Specifically, the following challenges have been identified:

(i) Regional Comparison Gap: Difficulty in comparing employment generation and work completion rates across different states and districts, making it challenging to identify high-performing and underperforming regions.

(ii) Lack of Trend Visibility: Absence of a consolidated view of yearly trends in employment, expenditure, and wages, limiting the ability to analyze growth patterns and policy impact over time.

(iii) Expenditure Opacity: Absence of clear insights into expenditure distribution across wages, material, and administrative costs, making it difficult to assess financial efficiency.

(iv) Limited Workforce Visibility: Limited visibility into workforce participation metrics such as number of individuals employed, households engaged, and average days of employment provided.

(v) No Unified District Platform: No unified platform to analyze district-level performance including total works taken up, total expenditure, and work completion rates.

(vi) Absence of Interactive BI Solution: Lack of an interactive dashboard providing a comprehensive real-time overview of MGNREGA performance using dynamic filters for year, state, and district.

In summary, the core problem lies in the absence of an integrated and interactive business intelligence solution capable of transforming raw MGNREGA data into meaningful, visual, and actionable insights for decision-makers. This project aims to address these challenges by developing a multi-page Power BI dashboard consolidating employment, expenditure, and performance metrics into a single user-friendly platform.

IV. OBJECTIVES OF THE PROJECT

The following six objectives were defined for this project, each aligned with a dedicated dashboard page developed in Power BI:

Objective 1 — Work Progress and Performance

Analysis: To analyze the overall progress of MGNREGA work activities by examining metrics such as total works undertaken, work completion rate, and average number of works across different years. This objective helps in understanding the efficiency and execution quality of the scheme at a macro level.

Objective 2 — State-wise Employment and Work Distribution Analysis:

To evaluate the distribution of works and employment generation across different states using map-based and comparative visualizations. This objective enables identification of high-performing states and regional disparities in implementation.

Objective 3 — Expenditure Analysis and Resource Allocation:

To analyze the distribution of total expenditure into key components such as wages, material costs, and administrative expenses. This objective helps assess financial efficiency and understand how funds are allocated under MGNREGA.

Objective 4 — Employment and Labour Participation

Analysis: To study workforce engagement by analyzing the number of individuals employed, households involved, and average days of employment provided. This objective highlights the effectiveness of MGNREGA in generating rural employment opportunities.

Objective 5 — District-Level Performance Monitoring:

To develop a detailed district-level dashboard presenting key performance indicators such as total works taken up, total expenditure, number of individuals employed, and completion rates. This objective allows for granular analysis and supports localized decision-making.

Objective 6 — Executive Summary and KPI Dashboard:

To design an interactive dashboard with key performance indicators supported by dynamic filters for year, state, and district. This objective provides decision-makers with a quick and comprehensive overview of

MGNREGA performance.

V. METHODOLOGY AND DATASET

A. Dataset Source

The dataset utilized in this project is derived from publicly available records of the Ministry of Rural Development, Government of India, pertaining to MGNREGA implementation [3]. The data is structured in CSV (Comma-Separated Values) format and encompasses region-wise and performance-based records related to employment generation, financial expenditure, and social-category participation. Each record represents aggregated data for a specific state or administrative unit over a defined financial year.

B. Dataset Description

The dataset includes the following key variables relevant to MGNREGA performance analysis:

Column Name	Data Type	Description
State	Categorical	Name of the state
District	Categorical	Name of the district (if available)
Financial Year	Categorical	Year of MGNREGA implementation
Total Households Worked	Numeric	Number of households that received employment
Total Person-Days	Numeric	Total number of workdays generated
SC Person-Days	Numeric	Person-days generated for Scheduled Castes
ST Person-Days	Numeric	Person-days generated for Scheduled Tribes
Women Person-Days	Numeric	Person-days generated for women workers
Total Wages Paid	Numeric	Total wages disbursed to workers
Total Expenditure	Numeric	Total funds spent under MGNREGA
Average Wage Rate	Numeric	Average wage paid per person-day

The dataset spans multiple financial years and includes records from all major states, providing a sufficiently large and varied foundation for multi-dimensional comparative analysis. Both employment and expenditure dimensions are covered, enabling comprehensive cross-sectional and longitudinal evaluation.

C. Data Preprocessing

Data preprocessing was conducted using Power Query in Power BI to ensure consistency, completeness, and

suitability for analysis. The following ten preprocessing steps were systematically applied:

Step	Action	Details
1	Removed Duplicate Records	Checked for duplicate entries based on state, district, and year; removed repeated rows to avoid double counting
2	Handled Null/Blank Values	Identified missing values in key columns; replaced nulls with zero or removed incomplete rows where necessary
3	Changed Data Types	Expenditure and wages set to Decimal; year to Whole Number; state/district to Text
4	Standardized Column Names	Renamed columns for consistency (e.g., Total_Exp → Total Expenditure, Avg_Wage → Average Wage)
5	Created Year Column	Ensured a proper Year column for time-based analysis and dynamic filtering
6	Created Calculated Columns	Derived average days of employment per household and work completion rate
7	Categorized Expenditure	Segregated total expenditure into wages, material costs, and administrative expenses
8	Aggregated Data	Grouped data at state and district levels for summary visualizations such as maps and bar charts
9	Removed Inconsistencies	Filtered out rows with unrealistic values (e.g., negative expenditure or zero employment records)
10	Verified Final Dataset	Conducted final validation for correct totals, consistent formatting, and accuracy

Following preprocessing, the dataset was enriched with derived columns including work completion rate and average employment days per household. The final clean dataset formed the analytical foundation for all dashboard pages described in the subsequent section.

VI. SYSTEM AND DASHBOARD DESIGN

The Power BI dashboard is structured as a multi-page interactive system with each page addressing a specific analytical objective. All pages incorporate interactive slicers for Year, State, and District, enabling dynamic data exploration without requiring technical expertise. DAX (Data Analysis Expressions) measures are used for computing KPIs such as total expenditure, average wage, and work completion rate. The following subsections describe each dashboard page, its components, and key observations.

A. Page 1: Rural Work Progress and Quality Dashboard

This page serves as the primary landing interface providing a macro-level overview of MGNREGA work progress and quality across different states and years. It is designed for policymakers, administrators, and analysts requiring a quick yet detailed understanding of scheme performance in

terms of work execution, employment generation, and resource utilization.

Key components include: (i) a combined column-and-line chart displaying average number of works undertaken alongside work completion rates across years; (ii) a choropleth map visualization of state-wise work distribution; (iii) a comparative bar chart contrasting jobs issued versus active jobs across states; (iv) a line chart tracking the percentage of Category B works over time; and (v) a stacked column chart depicting yearly expenditure distribution across wages, materials, and administrative costs.

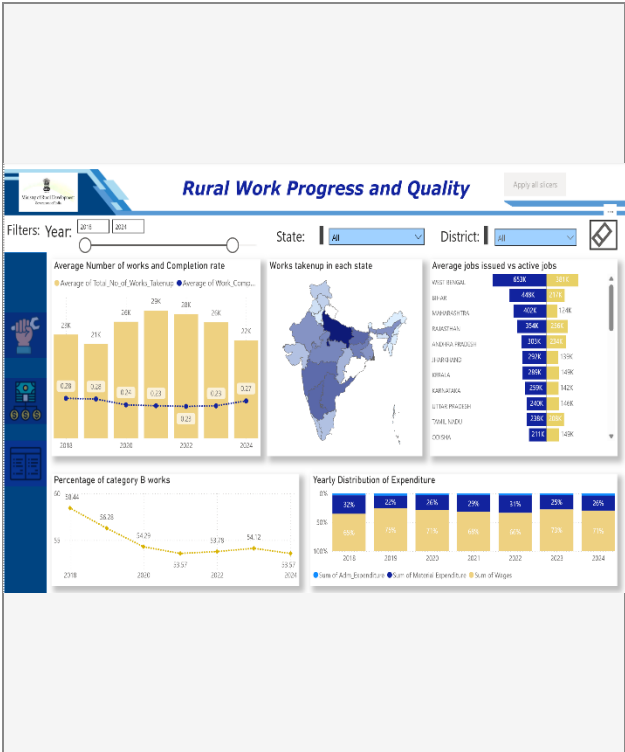


Fig. 1. Rural Work Progress and Quality Dashboard — Power BI Page 1

Key observations from Page 1: The number of works undertaken increased over the observed years before experiencing a slight decline in recent periods, while the work completion rate remained relatively stable, indicating consistent execution efficiency. The comparison between jobs issued and active jobs reveals a noticeable gap, suggesting that not all employment opportunities are fully utilized. The state-wise map highlights regional disparities, with certain states contributing significantly more to overall work activity. Expenditure distribution indicates that a major portion of funds is allocated to wages, reinforcing the labour-intensive nature of the scheme.

B. Page 2: Expenditure Information Dashboard

The second page focuses on the financial performance of the MGNREGA scheme by analyzing expenditure patterns and resource allocation across different states and years. It is designed for financial analysts, policymakers, and administrators who need to evaluate fund utilization effectiveness.

Key components include: (i) KPI cards for Total Expenditure, Total Individuals Employed, Material Expenses, and Average Wage; (ii) a bar chart displaying average individuals employed across states; (iii) a stacked bar chart showing percentage distribution of expenditure across wages, material costs, and administrative expenses; and (iv) a treemap visualization representing total expenditure by state for rapid identification of high-spending regions.

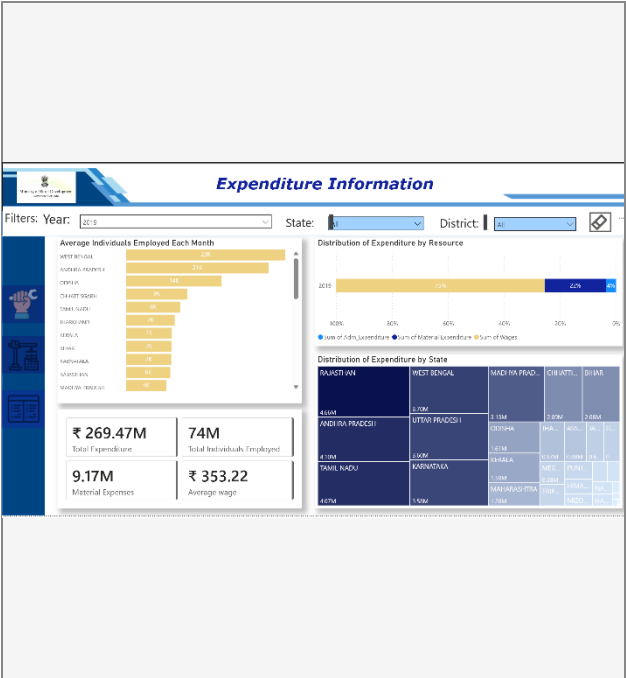


Fig. 2. Expenditure Information Dashboard — Power BI Page 2

Key observations from Page 2: A significant portion of total expenditure is consistently allocated towards wages, confirming the scheme's labour-oriented character. State-wise expenditure distribution shows noticeable variation, with certain states contributing a larger share of total spending, indicating regional differences in program scale and demand. Higher spending generally corresponds to increased employment generation; however, variations in average wage and material expenses suggest differences in cost efficiency and resource utilization across regions.

C. Page 3: Employment Infographics Dashboard

The third page provides detailed insights into employment generation and labour participation under MGNREGA. It

is designed to evaluate workforce engagement, identify high-performing regions, and assess whether the scheme is meeting its employment targets across states and districts.

Key components include: (i) a stacked area chart showing number of individuals employed across states over time; (ii) a bar chart highlighting top districts in terms of employment generation; (iii) a comparative chart displaying the relationship between average wages and total expenditure across districts; and (iv) a chart presenting average employment days per household alongside number of households engaged.

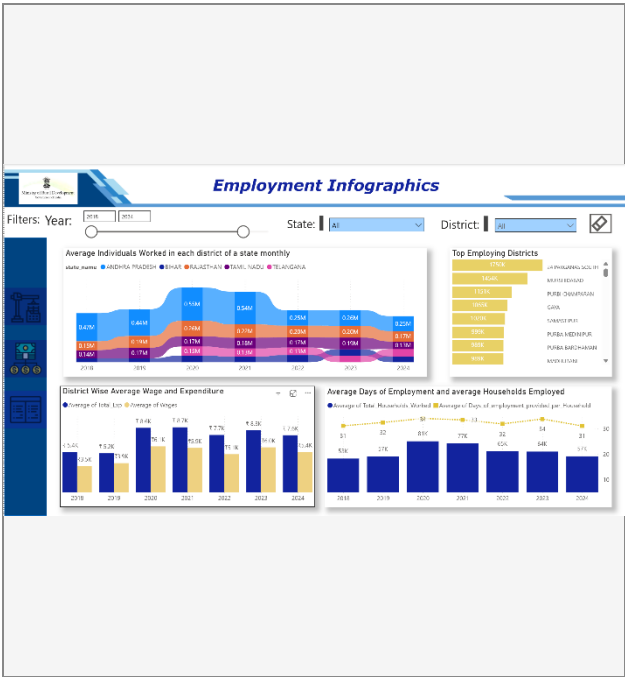


Fig. 3. Employment Infographics Dashboard — Power BI Page 3

Key observations from Page 3: Significant variation in employment generation is observed across states and districts. Certain regions consistently provide higher employment opportunities, indicating better implementation and higher demand for MGNREGA work. While a large number of individuals are employed under the scheme, the average number of employment days per household remains limited, suggesting the full 100-day employment guarantee is not being fully utilized. Wage patterns remain relatively stable while total expenditure varies depending on workforce scale.

D. Page 4: District Summary Dashboard

The fourth page provides a detailed summary of MGNREGA performance at the district level, offering granular insights into key performance indicators for specific districts. It is particularly useful for local administrators who need to monitor district-level

implementation and identify areas requiring improvement or additional support.

Key components include: (i) KPI cards for Work Completion Rate, Total Works Taken Up, Total Expenditure, and Individuals Employed; (ii) a structured district performance summary panel presenting key statistics for the selected district; and (iii) a contextual information panel with data source details and report context for enhanced usability.

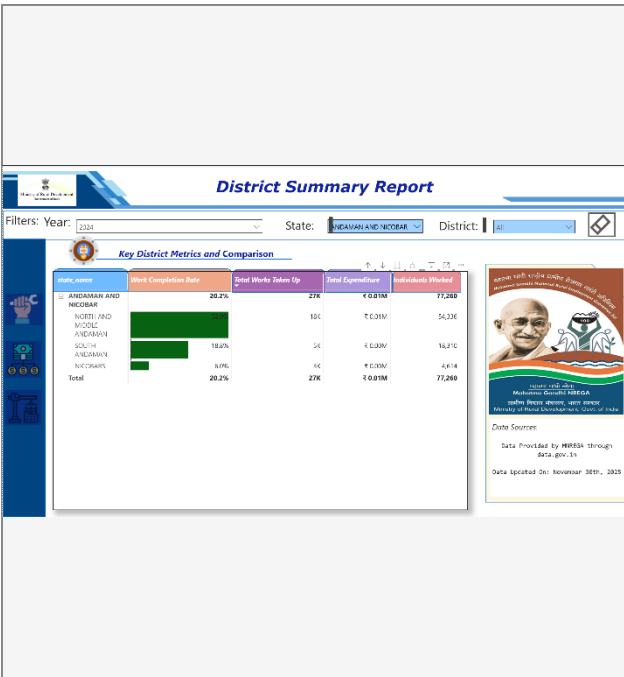


Fig. 4. District Summary Dashboard — Power BI Page 4

Key observations from Page 4: While a substantial number of works are undertaken and a large workforce is engaged at the district level, the work completion rate remains moderate in most districts, indicating room for improvement in execution efficiency. The comparison between total expenditure and employment figures suggests controlled spending relative to workforce scale, highlighting cost-effectiveness. Variations across districts indicate uneven implementation, enabling targeted identification of high-performing and underperforming areas.

E. Page 5: Executive Summary and KPI Dashboard

The fifth page serves as a high-level executive overview of overall MGNREGA performance. It consolidates the most critical KPIs—including total expenditure, total individuals employed, average wage, and material expenses—into a single interactive view supported by dynamic slicers for Year, State, and District. This page is designed to enable senior decision-makers to obtain a

rapid, data-driven assessment of scheme performance without navigating through detailed analytical pages.

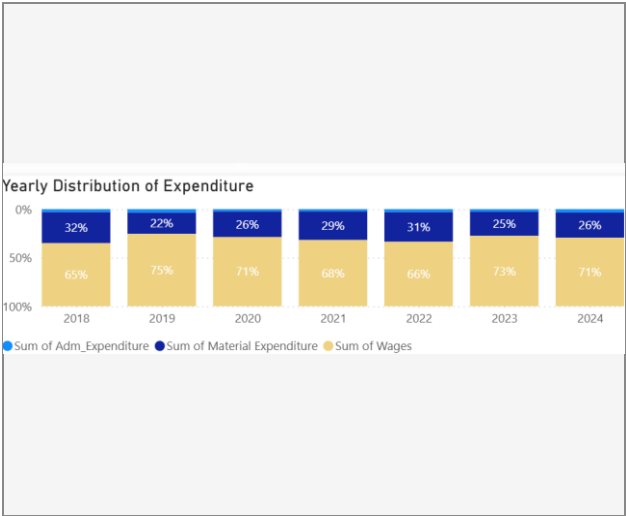


Fig. 5. Executive Summary and KPI Dashboard — Power BI Page 5

VII. RESULTS AND ANALYSIS

A. Work Progress Analysis

The work progress dashboard reveals that the number of MGNREGA works undertaken increased progressively over the observed years before a slight decline in recent periods. Work completion rates remained relatively stable throughout, indicating consistent execution efficiency. A notable gap between jobs issued and active jobs across multiple states suggests underutilization of available employment opportunities, pointing to potential issues in mobilization and demand fulfillment.

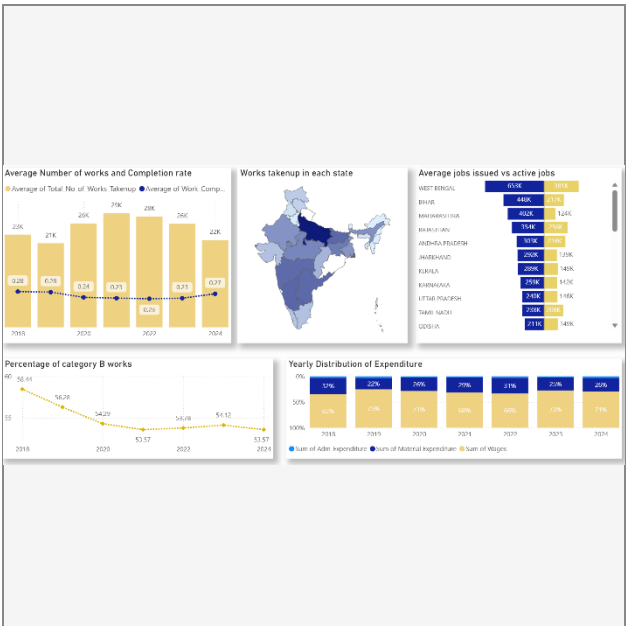


Fig. 6. Work completion rate trend and jobs issued vs. active jobs comparison

B. Expenditure and Resource Allocation Analysis

Expenditure analysis confirms that the largest proportion of total MGNREGA funds is consistently allocated to wages, affirming the labour-intensive character of the scheme. State-wise expenditure distribution exhibits significant variation, with certain states contributing disproportionately larger shares of total spending. The positive correlation between total expenditure and employment generation broadly holds; however, regional differences in average wage and material costs indicate variable cost-efficiency across administrative units.

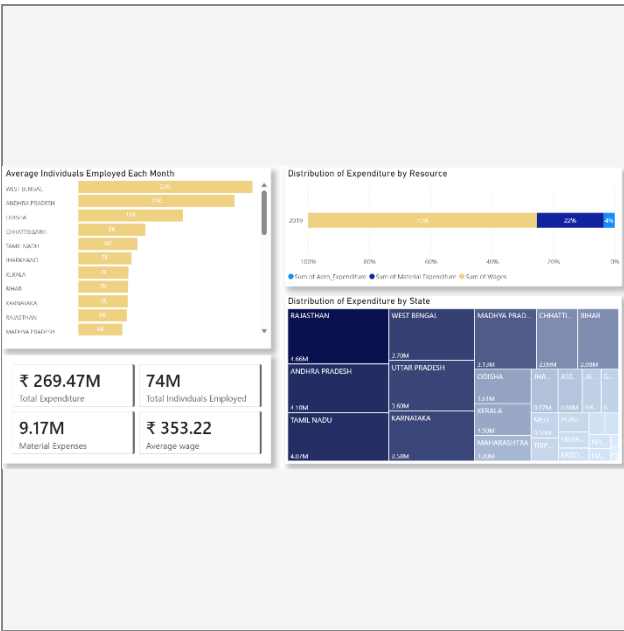


Fig. 7. State-wise expenditure distribution and expenditure component breakdown (wages, materials, admin)

C. Employment and Labour Participation

The employment dashboard reveals significant inter-state and inter-district variation in workforce engagement. While a large aggregate number of individuals are employed under the scheme, the average number of employment days per household falls substantially short of the stipulated 100-day guarantee in most regions, indicating only partial utilization of entitlement. Wage patterns remain relatively uniform, while total expenditure varies with the scale of workforce mobilization across regions.

D. District-Level Performance Findings

District-level analysis demonstrates that work completion rates vary considerably across administrative units. Certain districts demonstrate superior performance in both employment generation and expenditure utilization, while

others exhibit persistent underperformance. The district summary dashboard enables targeted identification of such disparities, providing the granular visibility needed for localized corrective interventions and resource reallocation decisions.

VIII. ADVANTAGES AND LIMITATIONS

A. Advantages

Centralized Data View: The dashboard integrates MGNREGA data from multiple dimensions—state, district, year, employment, and expenditure—into a single unified platform, eliminating the need to analyze multiple raw data files separately.

Interactive and Dynamic Filtering: Interactive slicers for Year, State, and District allow users to explore data dynamically without requiring technical expertise, making the dashboard accessible to both technical and non-technical stakeholders.

Multi-Level Analysis: The dashboard provides insights at national, state, and district levels, ensuring it meets the needs of policymakers, analysts, and local administrators alike.

Data-Driven Decision Making: By transforming raw MGNREGA data into meaningful visual insights, the dashboard supports evidence-based decision-making, helping authorities improve planning, monitoring, and execution of the scheme.

Identification of Underperforming Regions: State-level and district-level analysis enables quick identification of regions with low work completion rates, lower employment generation, or inefficient fund utilization, allowing timely corrective action.

Standardized KPIs: The use of calculated DAX measures and KPI cards ensures consistent computation of metrics such as total expenditure, employment levels, average wage, and completion rate, reducing discrepancies in analysis.

Clear and Intuitive Visualizations: Maps, bar charts, line charts, treemaps, and KPI cards ensure complex data is presented in a clear and visually intuitive manner, reducing interpretation effort.

Scalability and Flexibility: The Power BI model is scalable and can accommodate additional data from future years or new regions, ensuring long-term usability of the dashboard.

B. Limitations

Static Data Source: The current dashboard is connected to a static CSV/Excel dataset. It does not automatically update when new MGNREGA data becomes available, requiring manual data refresh or file replacement.

Single Data Source Dependency: The analysis is based solely on the MGNREGA dataset. The absence of integration with external data sources such as population data, economic indicators, or rural development indices limits the depth of insights.

No Predictive Analytics: The dashboard focuses on descriptive and diagnostic analysis only. It does not include forecasting models for predicting future employment trends or expenditure patterns, which would require integration with Python or R.

Limited Historical Data: The dataset covers a limited time period, which may not be sufficient for identifying long-term trends or evaluating the sustained impact of the scheme over multiple years.

No User Access Control: The dashboard does not implement row-level security (RLS), meaning all users have access to complete data across all states and districts, which may not be suitable in a real-world administrative environment.

Manual Data Preprocessing: The preprocessing steps in Power Query are dependent on the current dataset structure. Any changes in data format or additional fields would require manual adjustments in the transformation process.

Limited Analytical Scope: The dashboard primarily focuses on employment and expenditure metrics. Important aspects such as asset quality, long-term project impact, and beneficiary satisfaction are not captured.

No Mobile Optimization: Although Power BI supports mobile layouts, this dashboard has not been specifically optimized for mobile devices, which may affect usability on smaller screens.

IX. CONCLUSION

This paper presented the design, development, and evaluation of a comprehensive multi-page interactive analytics dashboard for monitoring MGNREGA performance using Microsoft Power BI. The system successfully addresses all six analytical objectives—covering work progress monitoring, state-wise distribution analysis, expenditure breakdown, employment and labour participation analysis, district-level performance evaluation, and executive KPI reporting—within a cohesive and user-friendly interface.

The analysis reveals important insights into MGNREGA implementation: regional disparities in employment generation and expenditure, gaps between jobs issued and actual employment utilization, the labour-intensive financial structure of the scheme with wages dominating total expenditure, and considerable variation in district-level work completion rates. From an academic perspective, this project provides hands-on experience with the complete data analytics workflow including data preprocessing, transformation, visualization, and interpretation.

In conclusion, this project illustrates how an interactive Power BI dashboard can transform raw government data into meaningful insights, supporting better decision-making, improved transparency, and more effective implementation of public welfare schemes.

The authors acknowledge the guidance of Maneet Kaur, School of Computer Science and Engineering, Lovely Professional University, whose supervision and constructive feedback shaped the direction and quality of this work.

X. FUTURE SCOPE

Several directions exist for extending this work. First, integration of Power BI with live government databases or cloud platforms would enable real-time monitoring of MGNREGA activities and eliminate the current static data limitation.

Second, embedding Python or R scripts within Power BI can facilitate predictive modeling for forecasting future employment demand, expenditure trends, and work completion rates, supporting proactive planning and resource allocation.

Third, incorporating external datasets—such as census population data, poverty indices, and rural development indicators—would enhance the depth of analysis and provide richer contextual insights into MGNREGA performance.

Fourth, Power BI's advanced AI features, including Key Influencers and Decomposition Tree visuals, can be leveraged to automatically identify factors affecting employment generation and expenditure patterns, enabling deeper analytical insights without manual investigation.

Fifth, migrating the data pipeline from CSV/Excel to a structured database system such as SQL Server or PostgreSQL would improve data management, enable automated updates, and support large-scale data handling. Publishing the dashboard to Power BI Service would

additionally enable web-based stakeholder access, scheduled refresh, and row-level security for role-based data access control.

Sixth, future versions can incorporate more detailed district-level metrics such as project types, duration, asset creation quality, and beneficiary satisfaction indices, providing deeper insights for localized decision-making and long-term impact assessment.

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12. LinkedIn

Project Link The project has been documented and shared on LinkedIn as part of the student's professional portfolio. The LinkedIn post includes a concise project overview, key screenshots of the Power BI dashboards, and a summary of the analytical objectives and insights derived from the MNREGA dataset. The purpose of sharing this project on LinkedIn is to showcase practical skills in data analysis, data visualization, and business intelligence using Power BI, as well as to engage with professionals and peers in the field of data science. LinkedIn Project Link:

[dhruvpowerbiproject](#)

Viewers are encouraged to visit the LinkedIn post to explore the dashboard visuals, understand the project methodology, and connect with the author for academic discussions or professional opportunities related to data analytics and business intelligence.